

Deviation register

Deviation register I

A registered report must state, explicitly, every way the executed analysis departed from the frozen plan. This template treats that register as a first-class artifact: `build_deviation_ledger` produces one row per executed outcome and model, classifying each as `ok` (registered), `warning` (an unregistered element carried with a documented rationale), or `error` (an unregistered element with no rationale). `compare_analysis_to_registration` downgrades a documented model change from `error` to `warning` and refuses to accept any deviation whose rationale is empty.

Demonstration deviations I

To exercise the machinery, the demonstration executes a deliberately plan-divergent analysis: it adds a `secondary_score` endpoint and swaps the registered `permutation_test` for a `linear_model`, each accompanied by a rationale. The resulting ledger is:

Order	Kind	Target	Registered	Documented	Severity
0	outcome	primary_score	yes	—	ok
1	outcome	secondary_score	no	yes	warning
2	model	linear_model	no	yes	warning

Demonstration deviations II

Because both departures are documented, the review packet remains valid: it reports `primary_score` as the only confirmatory outcome, `secondary_score` as exploratory, and an integrity score of 0.9 (a single 0.10 warning penalty against a perfect plan). Had either deviation lacked a rationale, the same code would have raised a blocking `deviation_without_rationale` or `unregistered_outcome` error, and the packet would be invalid. [`@fig:deviation_timeline`] plots this register along the registered-to-executed axis.

Demonstration deviations III

Preregistration-versus-executed deviation ledger timeline

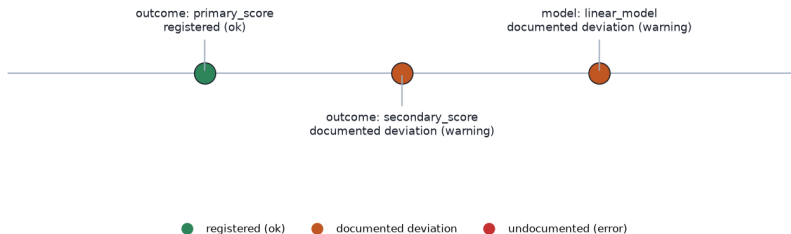


Figure 1: Preregistration-versus-executed deviation ledger timeline, rendered by `plot_deviation_timeline` from `build_deviation_ledger` and `deviation_timeline`. Each marker is a ledger row coloured by severity: the registered primary_score outcome is green (ok), while the added secondary_score endpoint and the linear_model substitution are amber (warning, documented deviations). No marker is red because every departure carries a rationale. An undocumented departure would appear in red and invalidate the review packet.

Demonstration deviations IV

Reading the register I

The register is the mechanism that keeps [Section @sec:results] honest: the confirmatory claim is restricted to the registered `primary_score` outcome, and the exploratory `secondary_score` endpoint is visible, labelled, and excluded from confirmatory interpretation. Forks should extend this ledger — never edit the frozen registration — whenever the executed analysis must diverge from plan.